Gear Fault Diagnosis in Geared Motors based on Frequency Adaptation Graph Prototype Network with Limited Data

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***Abstract*--Detecting and classifying gear faults with limited labeled data is essential for effective fault diagnosis in electromechanical systems. This paper presents a novel Frequency Adaptation Graph Prototype Network (FAGPN) for classifying various gear faults in geared motors. FAGPN employs customized low-pass and high-pass filters, integrated with an attention mechanism, to enhance multi-scale feature fusion from vibration signals processed through various frequency-time methods. Additionally, FAGPN projects embeddings onto hyperspherical space with a consistency constraint to improve accuracy. Experimental results on real-world datasets involving multiple gear sets in different states demonstrate the superior performance of FAGPN. Remarkably, with a training ratio of 1%, FAGPN achieves over 99% accuracy.**

Keywords—fault diagnosis, geared induction motor, graph neural network, limited data sample.

I. INTRODUCTION

Efficient condition monitoring and diagnosis of electric drives are essential for maintaining industrial electromechanical systems [1-4]. Existing predictive and diagnostic methods often require extensive prior experience and data to identify emerging issues, such as gear faults like tooth damage and wear , and various bearing faults (as shown in Fig. 1), in order to obtain a comprehensive analysis [5]. This necessity arises because data collected at different times is non-repeatable, and characteristic values representing machine conditions can vary within a certain range for the same model [6-8]. However, in real-world industrial settings, diagnosing faults is challenging due to datasets collected by sensors often being incomplete, imbalanced in fault classifications, and containing numerous unlabeled, noisy, or anomalous data. Therefore, it is crucial to develop methods that can accurately and promptly detect and prevent potential catastrophic failures, even with limited fault samples, to save both time and financial resources.

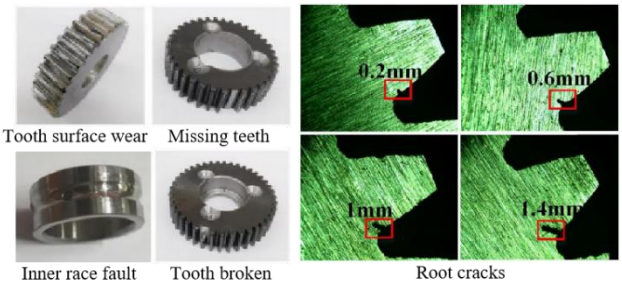


Fig.1. Examples of gear and bearing faults [5].

The most widely measured and studied fault detection indicator is unwanted mechanical vibration, typically caused by reciprocating forces, looseness, or resonance [9]. By manually or adaptively extracting and analyzing features from the time, frequency, and time-frequency domains, representative and distinguishable fault characteristics can be identified in data-driven diagnostic techniques for rolling electric drives. For example, phase demodulation in time-domain analysis, time-domain synchronous averaging using key phase signals for periodic truncation, power spectrum analysis in frequency-domain methods, high-resolution refined spectrum analysis, and cepstrum analysis for periodic components in the frequency domain [10-12]. Although these methods are effective, they heavily depend on expert knowledge for feature extraction and interpretation. To address the labor-intensive nature of traditional feature extraction methods, deep learning techniques have been extensively studied using vibration root mean square values, shape factor, crest factor, kurtosis factor, and other features. For instance, integrating a self-attention mechanism with convolutional neural networks (CNNs) [13], and incorporating graph neural networks (GNNs) to effectively manage data interdependencies for multi-scale feature fusion [14-16]. Nevertheless, these approaches generally require large datasets for optimal performance, which is challenging due to the scarcity of fault data relative to normal data in practice.

To address the challenge of small-sample fault classification for bearings and gearboxes, it is essential to develop effective methods. These methods could include improving the representativeness of features learned from limited data or employing transfer learning techniques [17-18]. In response to this challenge, we have developed a GNN designed to capture data interdependencies and dynamically aggregate multi-scale fault information from

This work was mainly supported by the funding support to State Key Laboratories in Hong Kong from the Innovation and Technology Commission (ITC) of the Hong Kong Government. Authors would also like to express their sincere thanks to the financial support from the Research and Innovation Office (Project code: P0046009 and BBX2) of The Hong Kong Polytechnic University (PolyU). This work was also supported in part by Start-up Fund for RAPs under the Strategic Hiring Scheme (P0043917) from PolyU and Research Committee of PolyU (Project code: RMCJ). (*Corresponding author: Dr. Zaixin Song; Email: zaixin.song@polyu.edu.hk)

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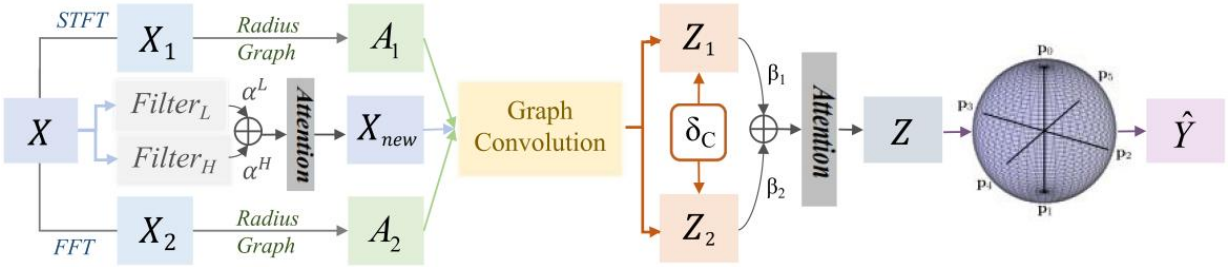


Fig.2. The workflow of the FAGPN model.

limited data. This frequency adaptation graph prototype network (FAGPN) utilizes customized low-pass and high-pass graph filters, integrated with an attention mechanism, to enhance multi-scale graph structure fusion through fast Fourier transform (FFT) and short-time Fourier transform (STFT). Furthermore, FAGPN projects the learned embeddings onto hyperspherical space with a consistency constraint to improve accuracy.

The remainder of this paper is organized as follows: Section II outlines the proposed methodology, Section III offers an overview of the dataset, comparative methods, and experimental results, and Section IV presents the conclusions of the study.

II. Frequency Adaptation Graph Prototype Network

This section presents the FAGPN model, which integrates graph structures derived from FFT and STFT and adaptively learns the relevant proportions to extract information pertinent to the node labels. Fig. 2 illustrates the workflow of the FAGPN model.

Our classification task is based on the feature matrix $X \in \mathbb{R}^{n\times d}$, which consists of vibration signals and the sample size, feature dimensionality are denoted as n, d, respectively. The feature matrix X is then transformed into X_1 and X_2 using STFT and FFT. Subsequently, the RadiusGraph method is employed to construct the feature graphs A_1 and A_2, where $A_i \in \mathbb{R}^{n\times n}, i = 1,2$ based on X_1 and X_2. Besides, the graph Laplacian matrix $L = I_n - D^{-1/2}AD^{-1/2}$ has eigenvectors $\{u_i\}_{i=1}^n \in \mathbb{R}^n$ and corresponding eigenvalues $\{\lambda_i\}_{i=1}^n \in [0,2]$, where $D \in \mathbb{R}^{n\times n}$ is the diagonal degree matrix and $I_n \in \mathbb{R}^{n\times n}$ is an identity matrix. The detailed methodology for obtaining the classification results is described as follows.

A. Adaptive Frequency-based Feature Extraction

To tackle fault classification in real-world scenarios, we use low-pass and high-pass graph filters to extract both low-frequency and high-frequency information. Integrating these types of information in varying proportions enables the model to handle situations where neighboring nodes may or may not share the same label.

By employing graph Fourier transform with tailored convolutional kernels g_θ, we can transform a signal x into the frequency domain using $\{u_i\}_{i=1}^n$. The filtered signal y can be represented as:

$$y = Ug_\theta(\Lambda)U^T x, \tag{1}$$

where U is the matrix of $\{u_i\}_{i=1}^n$, Λ is the diagonal matrix of eigenvalues $\{\lambda_i\}_{i=1}^n$, and $g_\theta(\Lambda)$ is a diagonal matrix with the filter $g_\theta(\lambda_i)$ applied to each λ_i.

Furthermore, we can separate the low-frequency and high-frequency graph signals into multi-scale frequency domains by defining two convolutional kernels $g_L(\lambda_i)$ and $g_H(\lambda_i)$ as follows [18]:

$$g_L(\lambda_i) = (1 - \lambda_i + \varepsilon)^2, \quad g_H(\lambda_i) = (\lambda_i - 1 + \varepsilon)^2, \tag{2}$$

where $\varepsilon \in [0,1]$ is an adjustment parameter.

These operations smooth the input signal (in the case of g_L) or emphasize the differences (in the case of g_H) between nodes based on their graph frequencies, facilitating the subsequent task. After employing the attention mechanism, the respective contributions of frequency components can be determined using α^L and α^H to represent their proportions.

B. Consistency-Driven Message Passing

Subsequently, we extract the embeddings Z_1, Z_2 from two graphs $G_1(A_1, X_{new})$ and $G_2(A_2, X_{new})$, where the updated feature matrix X_{new} originates from the aforementioned frequency adaptation component. Involving two graph convolution operations that run concurrently and share the same structure with parameter sharing, the process at the l-th layeris as follows:

$$Z_1^{(l)} = \mathrm{ReLU}(D_1^{-\frac{1}{2}}\widetilde{A_1}D_1^{-\frac{1}{2}}Z_1^{(l-1)}W^{(l)}), \tag{3}$$

$$Z_2^{(l)} = \mathrm{ReLU}(D_2^{-\frac{1}{2}}\widetilde{A_2}D_2^{-\frac{1}{2}}Z_1^{(l-1)}W^{(l)}), \tag{4}$$

where $\widetilde{A_1}$ and $\widetilde{A_2}$ are the adjacency matrices with self-loops, $Z_1^{(l-1)}$ and $Z_2^{(l-1)}$ are the embeddings from the previous layer (with $Z_1^{(0)} = Z_2^{(0)} = X_{new}$), and $W^{(l)}$ is the common weight matrix used in the l-th layer.

To enhance the consistency of information extraction across various spatial representations, we introduce a consistency constraint. Specifically, let $Z_i \bullet Z_i^T$ for $i = 1,2$ represent the similarity matrices [20]. The aim of this constraint is to minimize the divergence between these matrices, thereby enhancing the similarity and coherence of the extracted common information. The corresponding formula $\mathcal{L}_c$ is illustrated below :

$$\mathcal{L}_c = \|Z_1 \bullet Z_1^T - Z_2 \bullet Z_2^T\|_F^2, \tag{5}$$

where $\| \bullet \|_F^2$ is the squared Frobenius norm of a matrix.

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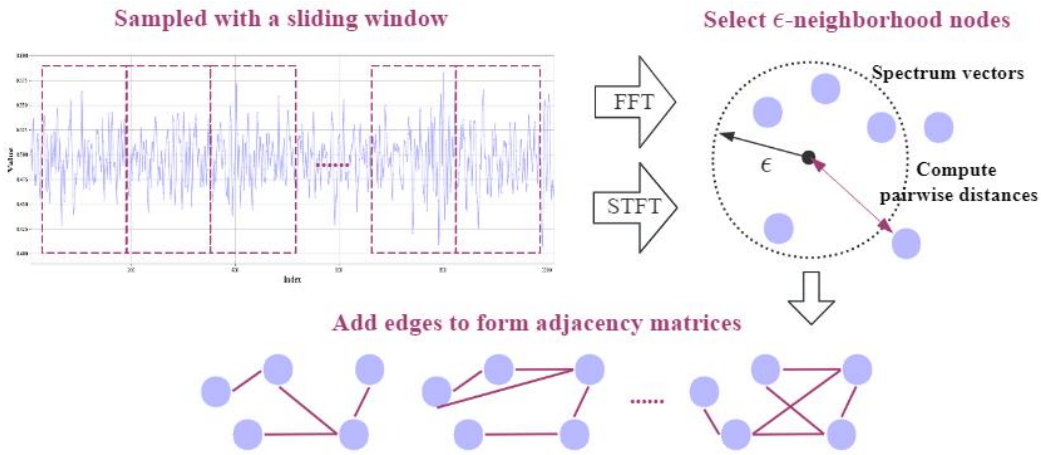


Fig.3. Graph construction using the RadiusGraph method.

TABLE I
STATISTICS OF THE DATASETS.

Dataset	Driving motor	Fault types	Class
XJTU Gearbox	3 Phase and 3 HP motor	Normal state (1) Bearing failure modes (4) Planetary gear failure modes (4)	9
XJTU Spurgear	AC variable frequency motor	Under 2 working conditions: Normal states (2) Spur gear failure modes (8)	10

Considering that the classification results might be linked to one or a combination of factors, we employ the attention mechanism to determine their respective importance through weights β_1 and β_2. It aims to extract the most relevant information, denoted as Z.

To ensure the classifier is effective for few-shot classification with limited examples of new classes, we employ hyperspheres as output spaces [21]. This ensures that the embedding matrix Z gradually align with class prototypes $p_i\ (i = 1, \ldots, N)$ obtained by unifying the output space and utilizing its positive portion prior to learning, rather than relying on one-hot vectors. Here, N is the number of categories.

Accordingly, the complete loss function comprises the consistency loss $\mathcal{L}_c$, the hyperspherical output space $\mathcal{L}_h$, and the cross-entropy loss $\mathcal{L}_e$. The overall loss function with adaptive weights can be expressed as:

$$\mathcal{L} = \gamma_1 \mathcal{L}_c + \gamma_2 \mathcal{L}_h + \gamma_3 \mathcal{L}_e, \tag{6}$$

where $\gamma_i\ (i = 1,2,3)$ are hyperparameters.

III. EXPERIMENTS

In this section, we employ real-world benchmark datasets with limited training samples to evaluate the effectiveness of the proposed FAGPN. Additionally, we present the results along with corresponding discussions to provide a comprehensive analysis.

A. Datasets

We focuses on a motor-gearbox system consisting of an electrical component (the geared induction motor) and a mechanical component (the gearbox). Uneven gear load, irregular tooth spacing, and load fluctuations caused by faults are reflected in vibration amplitude. The objective is to detect and classify failures in spur gears and planetary gears using vibration signals from multiple sensors. Datasets from XJTUGearbox and XJTUSpurgear were selected for the experiments [5]. Details about the datasets are provided in Table I.

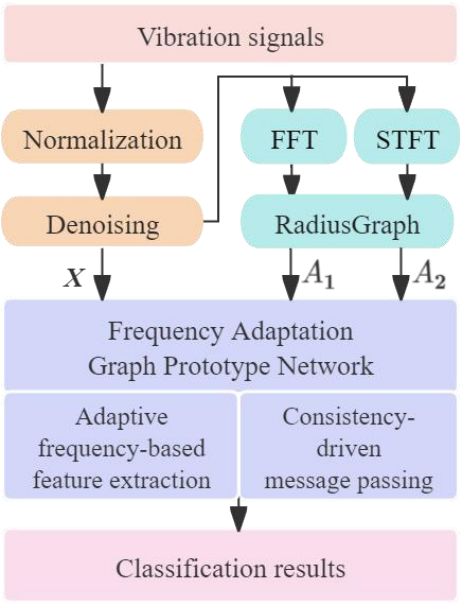


Fig. 4. Classification workflow based on the proposed FAGPN.

The XJTUGearbox dataset includes four planetary gear failure modes: tooth surface wear, missing tooth, root cracks, and broken tooth. Besides, it encompasses four bearing failure modes: ball fault, inner race fault, outer race fault, and a mixed fault involving the three aforementioned bearing faults. The XJTUSpurgear dataset contains four types of root cracks on the spur gear with crack lengths of 0.2 mm, 0.4 mm, 1 mm, and 1.4 mm.

B. Baselines

Several baseline GNN models in fault diagnosis, such as GCN, ChebyNet, SGCN, GAT, and GIN, are chosen

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TABLE III
ACCURACY COMPARISON OF TWO DATASETS.

Datasets	Train Ratio	GCN	ChebyNet	SGCN	GAT	GIN	FAGPN
XJTU Spurgear	80%	0.9913	0.9961	0.9940	0.9990	0.9970	**1.0000**
	50%	0.9885	0.8760	0.9321	0.9970	0.9911	**1.0000**
	10%	0.9733	0.8667	0.8703	0.9901	0.9158	**1.0000**
	5%	0.9441	0.7040	0.7984	0.9667	0.9049	**1.0000**
	1%	0.9366	0.6708	0.7640	0.9590	0.8981	**0.9828**
XJTU Gearbox	80%	**1.0000**	**1.0000**	**1.0000**	**1.0000**	0.9992	**1.0000**
	50%	**1.0000**	0.9396	0.9999	**1.0000**	0.9972	**1.0000**
	10%	0.9120	0.9091	0.9119	0.9073	0.8957	**1.0000**
	5%	0.9140	0.7906	0.9114	0.8709	0.9049	**1.0000**
	1%	0.9071	0.7008	0.8986	0.9007	0.8992	**1.0000**

TABLE II
NODE DISTRIBUTION IN TWO DATASETS.

Dataset	Train Ratio	Training Nodes	Validation Nodes	Testing Nodes
XJTU Gearbox	80%	8000	100	1000
	50%	5000		
	10%	1000		
	5%	500		
	1%	100		
XJTU Spurgear	80%	7200	90	900
	50%	4500		
	10%	900		
	5%	450		
	1%	90		

for comparison [22-25].

- GCN uses spectral graph convolutions to aggregate local node features effectively from neighboring nodes by stacking multiple convolutional layers.
- ChebyNet approximates spectral graph convolutions using Chebyshev polynomials, allowing it to handle larger neighborhoods with fewer parameters.
- SGCN reduces the complexity of GCN by collapsing multiple graph convolution layers into a single linear transformation followed by neighborhood aggregation.
- GAT introduces attention mechanisms to assign different weights to nodes in the neighborhood, enabling the model to focus on more relevant nodes.
- GIN distinguishes different graph structures more effectively than other GNNs by using sum aggregation followed by a multi-layer perceptron.

All models are built in a hardware environment using Python 3.8 and the PyTorch 1.10 framework on an Intel Xeon Platinum 8255C CPU @ 2.50GHz and an RTX 2080 Ti GPU. For each method, we ensure that the implementations align as closely as possible with those provided in the original papers, such as the learning rate is set to 0.004, weight decay to 10^{-4}, epoch to 1000 and dropout to 0.5. Additionally, all methods use average accuracy as the metric across 10 runs.

C. Data preprocessing

The preprocessing steps for time-series vibration data are as follows. Initially, the raw signals undergo Min-max normalization and wavelet decomposition and reconstruction for denoising, creating the feature matrix X for network input. Subsequently, the data is transformed using FFT to obtain the amplitude spectrum as a new feature matrix X_1 and STFT to derive the power spectral density as another feature matrix X_2. The transformed data is then continuously sampled using a sliding window with a length of 1024 and no overlap.

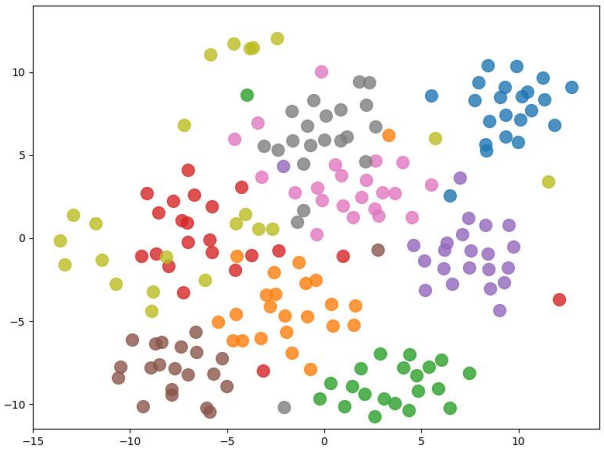
The resulting subsamples are processed using the RadiusGraph method, which creates connections between nodes within a certain radius to construct graph data. Fig. 3 illustrates the process of constructing graph structures using the RadiusGraph method. Additionally, Fig. 4 presents a flowchart detailing the entire process from data preprocessing to network training and the final classification results.

IV. RESULTS AND DISCUSSION

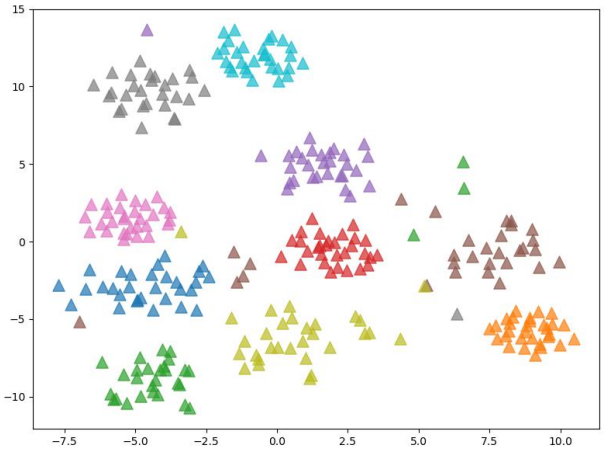
To conduct a thorough assessment of the classification task, we utilize consistent splits for the training, validation, and testing sets across the XJTUSpurgear dataset, which contains 9,000 nodes, and the XJTUGearbox dataset, which contains 1,0000 nodes. The training sets are divided into 1%, 5%, and 10% for performance validation under limited data scenarios, and 50% and 80% to represent traditional training set proportions. The validation set consists of 1% of the total nodes, while the testing set contains 10%. Table II presents the number of nodes in the training, validation, and testing sets for each dataset under different proportions mentioned above.

Furthermore, Table III shows the average accuracy for each method with the top-performing results highlighted in bold and the second-best results underlined. It can be seen that when the training ratio is 50% and 80%, most models, including GCN, ChebyNet, SGCN, GAT, and GIN, perform well, with scores close to 1.000. This indicates a high level of model robustness and the capability of these methods to generalize well from a significant portion of the training data. Specifically, in the XJTUGearbox dataset, GCN, SGCN, and GAT achieve perfect scores at the 80% training ratio and maintain high scores at 50%. Additionally, Fig. 5 presents a visual representation utilizing the t-distributed stochastic neighbor embedding (T-SNE) algorithm, which further elucidates the separability of embeddings within the XJTUGearbox and XJTUSpurgear datasets [26]. This visualization complements the model's performance by

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(a) XJTUGearbox



(b) XJTUSpurgear

Fig. 5. T-SNE visualization results of learned embeddings on (a) XJTUGearbox and (b) XJTUSpurgear.

demonstrating its ability to effectively capture and distinguish between different classes in the feature space.

However, as the training ratio decreases to 10%, 5%, and 1%, models such as ChebyNet and SGCN begin to struggle with the constraints of limited training data. For example, ChebyNet's performance on the XJTU Spurgear dataset declines significantly, dropping from 0.876 at the 50% training ratio to 0.704 at 5%, and further deteriorating to 0.671 at 1%. Fig. 6 provides a visual representation of this trend, clearly illustrating the performance retention of various methods at different training ratios through line graphs. The figures highlight how the models' abilities to maintain accuracy weaken as the amount of available training data diminishes.

This performance degradation can largely be attributed to the inherent mechanisms by which these models aggregate and propagate information. ChebyNet, which depends heavily on spectral methods, exhibits sensitivity to the quantity of training data, making it less capable of generalizing effectively when faced with sparse data. The spectral filtering process, central to ChebyNet's operation, becomes less effective with limited data, leading to a sharp decline in performance. Similarly, SGCN, designed to simplify the convolution process, may inadvertently lose crucial information during propagation, particularly

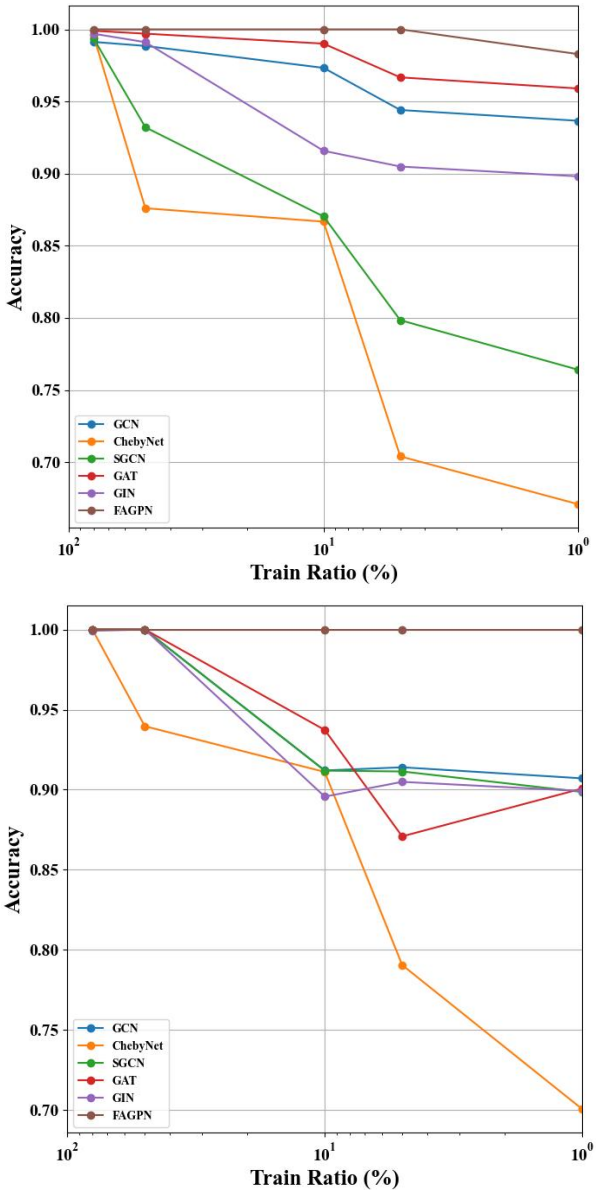


Fig. 6. Accuracy comparison on XJTU Spurgear and XJTU Gearbox.

when the training data is insufficient. This simplification, while beneficial in certain scenarios, proves to be a disadvantage under conditions of data scarcity. GAT and GIN, although more robust, still show performance drops, indicating challenges with limited data. In contrast, FAGPN consistently delivers high performance across all training ratios, demonstrating its sophisticated aggregation mechanism and better feature extraction capabilities, allowing it to use available information effectively even with scarce training data.

V. Conclusions

This paper introduces a novel FAGPN for diagnosing gear faults in geared motors, particularly under conditions of limited data. The proposed FAGPN is designed to effectively learn and prioritize the most relevant information by utilizing a combination of multi-scale filters, hyperspherical prototype projection, and a consistency constraint. FAGPN employs customized low-

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pass and high-pass filters specifically tailored to enhance the fusion of multi-scale features. These features are extracted from vibration signals processed using FFT and STFT techniques. By integrating these filters with an attention mechanism, the model focuses on the most informative aspects of the data, thereby improving the accuracy and reliability of fault diagnosis. Experimental results demonstrate FAGPN's effectiveness in scenarios characterized by data scarcity and imbalance. Remarkably, the model maintains high accuracy even when the available training data is extremely limited. Notably, when the training ratio is reduced to just 1%, FAGPN still achieves over 99% accuracy on the XJTUSpurgear and XJTUGearbox datasets. Future research may focus on incorporating additional domain-specific features, extending the model to other types of mechanical faults, and addressing complex compound fault scenarios.

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